

SEQUENTIAL POINT AND INTERVAL ESTIMATION OF THE SCALE PARAMETER IN A POWER FAMILY OF DISTRIBUTIONS WITH UNKNOWN SHAPE PARAMETER

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Summary: The approach of Mukhopadhyay, *et al.* (1983) is extended to provide a sequential estimation procedure for the scale parameter of a power family of distributions with unknown shape parameter. Asymptotic properties of the procedure are established and results of simulation studies (indicating the practical merits of the procedure for moderate sample sizes) are provided.

Sequential fixed-width confidence intervals for the scale parameter of the power family with unknown shape parameter are also developed. The procedure is shown to be "asymptotically efficient and consistent" in the sense of Chow and Robbins (1965). Simulation studies exhibit the utility of the procedure.

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1. INTRODUCTION

In this paper we focus on sequential and fixed width point estimation for the scale parameter in the power family of distributions with probability density functions (p.d.f.'s)

$$(1.1) \quad f_{\alpha, \theta}(x) = \frac{\alpha x^{\alpha-1}}{\theta^\alpha} I(0 \leq x \leq \theta), \quad \alpha, \theta > 0$$

where $I(\cdot)$ denotes the usual indicator function. Note that with $\alpha = 1$, (1.1) reduces to the (symmetric) uniform distribution over the interval $(0, \theta)$. Densities over the region $(0, \theta)$ which are skewed left (right) are obtained when α is greater (less) than one.

Several authors have presented sequential inferential procedures for θ when the distribution is uniform $(0, \theta)$. Graybill and Connell (1964) presented a two-stage procedure for construction of fixed-width confidence intervals for θ . Ghosh and Mukhopadhyay (1975) used as loss function absolute error plus cost of sampling in their derivation of a sequential point estimation procedure for θ . Using loss function squared error plus cost of sampling, Alvo (1978) discussed the same problem using a Bayesian approach.

Mukhopadhyay, *et al.* (1983) dealt with sequential and two-stage point estimation for θ in a power family distribution with known but arbitrary value for the shape parameter α . In this report, we focus on sequential point and fixed-width interval estimation for θ in the case where α is not assumed to be known but is a nuisance parameter.

If $X_1, X_2, \dots$ are i.i.d. with p.d.f. given by (1.1), the maximum likelihood estimators of θ and α^{-1} based on $X_1, \dots, X_n$ are given by

$\hat{\theta} = T_n = \max\{X_1, \dots, X_n\}$ and $\hat{\alpha}^{-1} = V_n = n^{-1} \sum_{i=1}^n (-\ln X_i + \ln T_n)$. Direct calculations show that T_n has p.d.f. $\theta^{-n\alpha} n^{\alpha-1} I(0 \leq t \leq \theta)$ and since $Y_i = -\ln X_i$ has p.d.f. $\{\alpha \exp(-\alpha(y + \ln \theta))\} I(y > -\ln \theta)$, it follows that

$2\alpha n V_n$ has a chi-square distribution with $2(n-1)$ degrees of freedom. Further, by Basu's Theorem, T_n and V_n are independent.

In Section 2 we consider the problem of estimating θ with minimum risk using the same loss function as Mukhopadhyay, *et al.* (1983). The sequential procedure is "asymptotically efficient", in a sense formally presented in Section 2. Section 3 is devoted to the development of a sequential fixed width interval estimation procedure for θ . It is shown that the proposed procedure is "asymptotically consistent" and "asymptotically efficient" in the sense of Chow and Robbins (1965). Monte Carlo simulations of the point estimation and interval estimation procedures are presented in Section 4. Generally, the procedures perform quite well.

Throughout the report we make use of the above-mentioned distributional properties and the following notations:

$[x]$ denotes the greatest integer $\leq x$; for positive a , $\Gamma(a)$ denotes the gamma function $\int_0^\infty e^{-x} x^{a-1} dx$; and $a(y) \approx b(y)$ denotes that $a(y)/b(y) \rightarrow 1$ as $y \rightarrow \infty$.

2. SEQUENTIAL POINT ESTIMATION OF θ

Let $X_1, X_2, \dots$ be i.i.d. with density given in (1.1). Having observed $X_1, \dots, X_n$, let $T_n = \max\{X_1, \dots, X_n\}$ and consider estimating θ by T_n with loss

$$(2.1) \quad L_n = A(\theta - T_n)^s + cn,$$

where $A(>0)$ is a known weight, $s(>0)$ is known and $c(>0)$ is the known cost per unit sample. As in Mukhopadhyay, *et al.* (1983), the risk associated with (2.1) is

$$(2.2) \quad \nu_n(c) = E(L_n) \approx A\theta^s \Gamma(s+1)(n\alpha)^{-s} + cn$$

and if $d = sA\Gamma(s+1)$

$$(2.3) \quad n^* = (d\theta^s / (c\alpha^s))^{1/(s+1)}$$

is the value of n which minimizes (2.2). Note that $\nu_{n^*}(\hat{c}) \approx cn^*(1+s^{-1})$ and that since θ and α are unknown, n^* given in (2.3) cannot be determined. We propose the following "delayed" estimation procedure. Let $m \geq 2$ be the initial sample size and for some fixed $\gamma > 0$ define

$$(2.4) \quad N = N(c) = \inf\{n \geq m \text{ such that } n \geq (d/c)^{1/(s+1)} ((T_n \nu_n)^{s/(s+1)} + n^{-\gamma})\}.$$

The "delay" factor of $n^{-\gamma}$ is used to prevent very early stopping and the choice of γ reflects the amount of delay built into the procedure. When the rule given in (2.4) indicates that sampling be terminated, estimate θ by T_N . Note that this procedure differs from many others presented in the literature in that the stopping rule, N , and the estimator are *not* independent of one another. The properties of the procedure are summarized in the following theorem.

Theorem 1. If $0 < \alpha < \infty$, $0 < \theta < \infty$, and $0 < \gamma < \infty$, the following results hold for the procedure proposed in (2.4):

- (i) N is well defined and non-increasing in c ;
- (ii) $N \rightarrow \infty$ a.s. as $c \rightarrow 0$;
- (iii) $N/n^* \rightarrow 1$ a.s. as $c \rightarrow 0$;
- (iv) $E(N)/n^* \rightarrow 1$ as $c \rightarrow 0$;
- (v) If $\nu(c) = AE\{(\theta - T_N)^s\} + cE(N)$ then
 - $\nu(c)/\nu_{n^*}(c) \rightarrow 1$ as $c \rightarrow 0$ (asymptotic risk efficiency).

Proof: The proofs of (i) and (ii) are immediate from (2.4). Also (2.4) implies that

$$(2.5) \quad (d/c)^{1/(s+1)} (T_N V_N)^{s/(s+1)} \leq N \\ \leq m + (d/c)^{1/(s+1)} \{(T_{N-1} V_{N-1})^{s/(s+1)}\} + (N-1)^{-\gamma}.$$

Since $N \rightarrow \infty$, a.s. as $c \rightarrow 0$, $T_N \rightarrow \theta$ a.s. and $V_N \rightarrow \alpha^{-1}$ a.s..

Dividing all terms in (2.5) by n^* yields (iii).

To prove (iv), write $Y_i = -\ln X_i$ and note that $T_n \leq \theta$ a.s. for all n .

Thus, $V_n = n^{-1} \sum_{i=1}^n (Y_i + \ln T_n) \leq n^{-1} \sum_{i=1}^n (Y_i + \ln \theta)$ a.s.

Also, (2.4) implies that $N \geq (d/c)^{1/((s+1)(\gamma+1))}$ and, hence, that for small enough c , $N-1 > 1/2 (d/c)^{1/((s+1)(\gamma+1))}$. Further, since $Y_i + \ln \theta \geq 0$ for all i , it follows that

$$N-1 \leq m-1 + (d/c)^{1/(s+1)} \{\theta (N-1)^{-1} \sum_{i=1}^N (Y_i + \ln \theta)\}^{s/(s+1)} \\ + 2^\gamma (d/c)^{1/((s+1)(\gamma+1))}.$$

Multiplying the above inequality through by $(N-1)^{s/(s+1)}$ and applying the Hölder inequality yields that

$$\{E(N-1)\}^{(2s+1)/(s+1)} \\ \leq (m-1) \{E(N-1)\}^{s/(s+1)} + (d\theta^s/c)^{1/(s+1)} \{E(\sum_{i=1}^N (Y_i + \ln \theta))\}^{s/(s+1)} \\ + 2^\gamma (d/c)^{1/((s+1)(\gamma+1))} \{E(N-1)\}^{s/(s+1)}.$$

If $E(N) < \infty$, since $E(Y_i + \ln \theta) = \alpha^{-1}$, Wald's Lemma implies that

$$\{E(N-1)\}^{(2s+1)/(s+1)} \\ \leq (m-1) \{E(N-1)\}^{s/(s+1)} + (d\theta^s/c)^{1/(s+1)} \{\alpha^{-1} E(N)\}^{s/(s+1)} + \\ + 2^\gamma (d/c)^{1/((s+1)(\gamma+1))} \{E(N-1)\}^{s/(s+1)}.$$

Dividing through by $\{E(N-1)\}^{s/(s+1)}$ and recalling that $n^* = (d\theta^s/(\alpha s))^{1/(s+1)}$ we obtain

$$E(N-1) \leq m-1 + n^* (E(N)/E(N-1))^{s/(s+1)} + 2\gamma (\alpha/\theta)^s / ((s+1)(\gamma+1)) n^{*1/(\gamma+1)}.$$

If we do not assume $E(N) < \infty$, define $N_j = \min\{N, j\}$ and apply the above technique to show that $E(N_j) \leq m + n^* (E(N_j)/E(N_j-1))^{s/(s+1)} + K n^{*1/(\gamma+1)}$.

Since N_j increases to N the monotone convergence theorem gives the desired result.

Since $E(N) \rightarrow \infty$ as $c \rightarrow 0$, it follows that $\limsup E(N/n^*) \leq 1$ as $c \rightarrow 0$. Fatou's theorem and (iii) imply that $\liminf E(N/n^*) \geq 1$ as $c \rightarrow 0$. Thus (iv) follows immediately.

The proof of (v) is more involved. Note that

$$\nu(c)/\nu_{n^*}(c) \approx \{AE(\theta - T_N)^s + cE(N)\} (cn^*(1+s^{-1}))^{-1}.$$

Since $E(N/n^*) \rightarrow 1$, as $c \rightarrow 0$, to establish (v) we need only show that $AE(\theta - T_N)^s (cn^*)^{-1} \rightarrow 1$ as $c \rightarrow 0$. Throughout the remainder of the proof we treat n^* as an integer. It is easily seen that

$AE(\theta - T_{n^*})^s (cn^*)^{-1} \rightarrow 1$ as $c \rightarrow 0$, so it is sufficient to show that

$$E\{(\theta - T_N)^s - (\theta - T_{n^*})^s\} (cn^*)^{-1} \rightarrow 0 \text{ as } c \rightarrow 0.$$

Note that from (2.4) $N \geq (d/c)^{1/(s+1)} N^{-\gamma}$ and, hence, that

$N \geq (d/c)^{1/((s+1)(\gamma+1))}$. Choose $\eta \in (0, 1)$, let $\varepsilon = 1 - \eta$ define $u =$ smallest integer $\geq (d/c)^{1/((s+1)(\gamma+1))}$. Write

$$(2.6) \quad |(\theta - T_N)^s - (\theta - T_{n^*})^s| = |(\theta - T_N)^s - (\theta - T_{n^*})^s| \{I(N \leq [\varepsilon n^*]) + I([\varepsilon n^*] < N \leq n^*) \\ + I(n^* < N \leq [(1+\eta)n^*]) + I(N > [(1+\eta)n^*])\}.$$

Let us look at the first term on the RHS of (2.6). Note that

$$|(\theta - T_N)^S - (\theta - T_{n^*})^S| \leq 2\theta^S \text{ and therefore}$$

$$(2.7) \quad E\{ |(\theta - T_N)^S - (\theta - T_{n^*})^S| I(N \leq [en^*]) \} \leq 2\theta^S P\{N \leq [en^*]\}.$$

It follows that

$$(2.8) \quad P\{N \leq [en^*]\} \leq \sum_{n=u}^{[en^*]} P(n \geq (d/c)^{1/(s+1)} (T_n V_n)^{s/(s+1)}) \\ = \sum_{n=u}^{[en^*]} P(T_n V_n \leq n^{(s+1)/s} (d/c)^{-1/s}).$$

$$\text{Let } W_n = T_n V_n \text{ and note that } E(W_n) = (E T_n)(E V_n) = \theta(n-1)(n\alpha+1)^{-1}.$$

Thus, subtracting $E(W_n)$ from both sides of the inequality in (2.8),

$$P\{N \leq [en^*]\} \leq \sum_{n=u}^{[en^*]} P(W_n - E(W_n) \leq n^{(s+1)/s} (d/c)^{-1/s} - \theta(n-1)(n\alpha+1)^{-1}) \\ \leq \sum_{n=u}^{[en^*]} P(W_n - E(W_n) \leq (en^*)^{(s+1)/s} (d/c)^{-1/s} - \alpha^{-1} \theta + \theta(1+\alpha)\{(n\alpha+1)\alpha\}^{-1}) \\ \leq \sum_{n=u}^{n^*} P(W_n - E(W_n) \leq \alpha^{-1} \theta (\epsilon^{(s+1)/s} - 1) + \theta(1+\alpha)\{(u\alpha+1)\alpha\}^{-1}).$$

Let $\delta = 1 - \epsilon^{(s+1)/s}$ and choose c small enough that

$$\theta(1+\alpha)\{(u\alpha+1)\alpha\}^{-1} \leq \delta\theta(2\alpha)^{-1}. \text{ Then}$$

$$P\{N \leq [en^*]\} \leq \sum_{n=u}^{n^*} P(W_n - E(W_n) \leq -\delta\theta(2\alpha)^{-1}) \\ \leq \sum_{n=u}^{n^*} P(|W_n - E(W_n)| \geq \delta\theta(2\alpha)^{-1}) \\ \leq \sum_{n=u}^{n^*} (\delta\theta(2\alpha)^{-1})^{-2r} E|W_n - E(W_n)|^{2r} \text{ for all } r > 0.$$

$$\text{Since } E|W_n - E(W_n)|^{2r} = E|T_n(V_n - EV_n) + EV_n(T_n - ET_n)|^{2r} \\ \leq 2^{2r} \{E|T_n(V_n - EV_n)|^{2r} + E|EV_n(T_n - ET_n)|^{2r}\} \\ \leq 2^{2r} (\theta^{2r} E|V_n - EV_n|^{2r} + |E(V_n)|^{2r} E|T_n - ET_n|^{2r}) \\ = O(n^{-r}),$$

it follows that for some constant k' , all $\varepsilon \in (0,1)$

$$P\{N \leq [cn^*]\} \leq k'(\delta)^{-2r} \sum_{n=u}^{n^*} n^{-r} \leq k'n^* u^{-r}.$$

Therefore, from (2.7), for all $\varepsilon \in (0,1)$

$$(2.9) \quad (cn^*)^{-1} E\{ |(\theta - T_n)^S - (\theta - T_{n^*})^S| I(N \leq [cn^*]) \} \\ \leq k'u^{-r} \varepsilon^{-1} \leq k'\varepsilon \{r/((s+1)(\gamma+1)) - 1\}.$$

Choosing $r > (s+1)(\gamma+1)$, the RHS of (2.9) $\rightarrow 0$.

Since T_n is an increasing function of n and $T_n \leq \theta$ a.s. for all $n \geq 1$, and $\varepsilon = 1-\eta$ the second term on the RHS of (2.6) is such that

$$|(\theta - T_n)^S - (\theta - T_{n^*})^S| I([(1-\eta)n^*] < N \leq n^*) \leq (\theta - T_{[n^* - \eta n^*]})^S - (\theta - T_{n^*})^S.$$

Therefore,

$$(2.10) \quad E(|(\theta - T_n)^S - (\theta - T_{n^*})^S| I([(1-\eta)n^*] < N \leq n^*)) \\ \leq \theta^S \Gamma(s+1) \left\{ \frac{\Gamma(n^*(1-\eta)\alpha+1)}{\Gamma(n^*(1-\eta)\alpha+s+1)} - \frac{\Gamma(n^*\alpha+1)}{\Gamma(n^*\alpha+s+1)} \right\}.$$

In an analogous manner

$$(2.11) \quad E(|(\theta - T_n)^S - (\theta - T_{n^*})^S| I(n^* < N \leq [(1+\eta)n^*])) \\ \leq \theta^S \Gamma(s+1) \left\{ \frac{\Gamma(n^*\alpha+1)}{\Gamma(n^*\alpha+s+1)} - \frac{\Gamma(n^*(1+\eta)\alpha+1)}{\Gamma(n^*(1+\eta)\alpha+s+1)} \right\}.$$

Thus, from (2.10) and (2.11),

$$(2.12) \quad (cn^*)^{-1} E\{ |(\theta - T_n)^S - (\theta - T_{n^*})^S| (I([(1-\eta)n^*] < N \leq n^*) + I(n^* < N \leq [(1+\eta)n^*])) \} \\ \leq (cn^*)^{-1} \theta^S \Gamma(s+1) \left\{ \frac{\Gamma(n^*(1-\eta)\alpha+1)}{\Gamma(n^*(1-\eta)\alpha+s+1)} - \frac{\Gamma(n^*(1+\eta)\alpha+1)}{\Gamma(n^*(1+\eta)\alpha+s+1)} \right\}.$$

Using upper and lower bounds for $\Gamma(x+1)$ as given in (6.1.38) of Abramowitz and Stegun (1970) it is not difficult to show that the bracketed term on the

RHS of (2.12) is bounded above by $(n^*\alpha)^{-S}G(\eta, c)$ where $G(\eta, c) \rightarrow (1-\eta)^{-S} - (1+\eta)^{-S}$ as $c \rightarrow 0$. Since $(1-\eta)^{-S} - (1+\eta)^{-S}$ can be made arbitrarily small by choosing η small, it follows that the RHS of (2.12) is bounded above by $\theta^S \Gamma(s+1) (\alpha^{-S} (n^*)^{-(s+1)} / c) G(\eta, c) \leq k^* G(\eta, c)$ for a constant k^* . Thus, the LHS of (2.12) can be made arbitrarily small by choosing η close to 0.

Finally, we look at

$$|(\theta - T_N)^S - (\theta - T_{n^*})^S| I(N > n^*(1+\eta)) = \{(\theta - T_{n^*})^S - (\theta - T_N)^S\} I(N > n^*(1+\eta)) \\ \leq (\theta - T_{n^*})^S I(N > n^*(1+\eta)).$$

Hölder's inequality implies that

$$(2.13) \quad (cn^*)^{-1} E\{|(\theta - T_N)^S - (\theta - T_{n^*})^S| I(N > n^*(1+\eta))\} \\ \leq (cn^*)^{-1} \{E(\theta - T_{n^*})^{2S}\}^{1/2} \{P(N > n^*(1+\eta))\}^{1/2}.$$

Since it is easily shown that $E(\theta - T_{n^*})^{2S} = O(n^{*-2S})$, and we know $N/n^* \rightarrow 1$ a.s., the RHS of (2.12) converges to zero as $c \rightarrow 0$. Thus, (vi) follows from (2.6), (2.9), (2.12) and (2.13).

The results given in Theorem 1 are asymptotic in nature and require the choice of a value of $\gamma > 0$. Although we have not been able to prove analogous results with no delay factor (i.e. $\gamma = \infty$ in (2.4)), the simulation results in Section 4 indicate that the rule without the delay factor performs poorly. Whereas, even for moderate c values, the rule addressed in Theorem 1 performs quite well.

3. SEQUENTIAL FIXED WIDTH INTERVAL ESTIMATION OF θ

In this section, our objective is to obtain a confidence interval for θ which is of length $d (> 0)$ and has prescribed confidence coefficient $1-\delta$ for $(0 < \delta < 1)$. Let $X_1, X_2, \dots$ be i.i.d. with p.d.f. as given in

(1.1) and T_n and V_n be the maximum likelihood estimators of θ and α^{-1} described in Section 1. We consider the confidence interval

$I_n = (T_n, T_n + d)$ for θ . If θ and α were known

$P_\theta(\theta \in I_n) = P(\theta - d \leq T_n \leq \theta) = 1 - ((\theta - d)/\theta)^{n\alpha} \geq 1 - \exp(-nad/\theta)$. In order that $P(\theta \in I_n) \geq 1 - \delta$ it suffices that $\exp(-nad/\theta) \leq \delta$. Equivalently, if

$$(3.1) \quad n^\# = -\theta \ln \delta / (d\alpha),$$

$n^\#$ is the sample size which achieves our objectives. However, since θ and α are unknown, no fixed sample size can achieve the objective for all θ, α . Let $Y_1 = -\ln X_1$, $m \geq 2$ denote an initial sample size, $\gamma > 0$ be a fixed constant and define

$$(3.2) \quad N = N(d) = \inf\{n \geq m \text{ such that } n \geq d^{-1}(-\ln \delta)(T_n V_n + n^{-\gamma})\}.$$

If sampling is terminated at stage N , use I_N as the confidence interval. Again, the introduction of γ is necessary to prevent very early stopping. Results very similar to those we present can be derived with no delay parameter (i.e., $\gamma = \infty$) but the simulation results presented in Section 4 indicate that finite choices of γ produce clearly superior results. For this reason, we confine our discussion to choices of $0 < \gamma < \infty$. The properties of the sequential confidence interval procedure are summarized in the following theorem.

Theorem 2. If $0 < \alpha < \infty$, $0 < \theta < \infty$, and $0 < \gamma < \infty$. The following results hold for the procedure proposed in (3.2).

- (i) N is well defined and non-increasing in d ;
- (ii) $N \uparrow \infty$ a.s. as $d \rightarrow 0$;
- (iii) $N/n^\# \rightarrow 1$ a.s. as $d \rightarrow 0$;
- (iv) $E(N)/n^\# \rightarrow 1$ as $d \rightarrow 0$;
- (v) $P(\theta \in I_N) \rightarrow 1 - \delta$ as $d \rightarrow 0$.

Proof: The proofs of (i) through (iv) are very similar to their counterparts in Theorem 1. We focus primarily on the proof of (v). Routine calculations yield that $Z_i = (X_i/\theta)^\alpha$ are i.i.d. uniform (0,1) random variables and that for all n , $-\alpha \ln(T_n/\theta)$ has an exponential distribution with mean 1. Further, Theorems 1 and 6 of Anscombe (1952) imply that, as $c \rightarrow 0$, $-\alpha \ln(T_n/\theta)$ has the same limiting exponential distribution. Now

$$\begin{aligned} P(\theta \in I_N) &= P\{\theta - d \leq T_N \leq \theta\} \\ &= P\{0 \leq -\alpha \ln(T_N/\theta) \leq -\alpha \ln((\theta - d)/\theta)\}. \end{aligned}$$

Note that $-\alpha \ln((\theta - d)/\theta) = \{-n^\# \alpha \ln((\theta - d)/\theta)\} N/n^\# \rightarrow -\ln \delta$ as $d \rightarrow 0$ by (iii) and the relationship $\ln((\theta - d)/\theta) \approx (d/\theta)$. Thus, the proof is complete.

In the next section we present the results of simulation studies which indicate that the proposed sequential confidence interval procedure performs quite well, especially for choices of γ near 1.

4. MONTE CARLO RESULTS

In this section we present the results of numerical studies implemented using FORTRAN IV programs on an IBM 3081D system. Section 4.1 contains the simulation results for the sequential point estimation procedure presented in Section 2. Analogous results for the sequential confidence interval procedure presented in Section 3 are contained in Section 4.2.

4.1 Simulation results for the point estimation procedure of Section 2

Throughout the simulations we considered the power family of densities (see (1.1)) with $\theta=1$ and $\alpha=1/2, 1, 3$. We used loss function (2.1) with $A=1$, and $s=2$, for which $n^*=(4(c\alpha^2)^{-1})^{1/3}$. In these cases, (2.2) yields that $\nu_n(c) \approx 3cn^*/2$. Rather than choosing various values

of the cost per observation, c , and determining the corresponding values of n^* , we focused on nicely spaced integer values of n^* and worked with values of c which yield the desired values for n^* . For $n^*=5, 25$ and 100, we computed $c=4/(\alpha^2 n^{*3})$ for the various values of α . Note that c is small when n^* is large (smaller when α is also large). The estimation procedure was applied 500 times for each choice of α and n^* . Each time sampling terminated, we determined the observed value of N , say $n(i)$, the observed value of T_N , say $T_{n(i)}$, V_N , say $V_{n(i)}$, and $L_{n(i)} = (1 - T_{n(i)})^2 + cn(i)$. Throughout the tables

$$\bar{n} = k^{-1} \sum_{i=1}^k n(i), \quad SD(n) = \{k^{-1} \sum_{i=1}^k (n(i) - \bar{n})^2\}^{1/2},$$

$$\bar{T} = k^{-1} \sum_{i=1}^k T_{n(i)}, \quad SD(T) = \{k^{-1} \sum_{i=1}^k (T_{n(i)} - \bar{T})^2\}^{1/2},$$

$$\bar{\alpha} = k^{-1} \sum_{i=1}^k V_{n(i)}^{-1} \quad \text{and} \quad SD(\hat{\alpha}) = \{k^{-1} \sum_{i=1}^k (V_{n(i)}^{-1} - \bar{\alpha})^2\}^{1/2}$$

$\bar{L} = k^{-1} \sum_{i=1}^k L_{n(i)}$ with $k = 500$. We consider $\bar{L}$ as an estimate of $\nu(c)$ (see Theorem 1) and estimate $\eta = \nu(c)/\nu_{n^*}(c)$ by $\hat{\eta} = \bar{L}/\nu_{n^*}(c)$.

Further, $\omega = \nu(c) - \nu_{n^*}(c)$ is estimated by $\hat{\omega} = \bar{L} - \nu_{n^*}(c)$. If the procedure performs well, we would expect that $\bar{T} \approx \theta = 1$, $\bar{n} \approx n^*$, $\bar{\alpha} \approx \alpha$ and that $n(i)$, $T_{n(i)}$ and $\hat{\alpha}_{n(i)} = V_{n(i)}^{-1}$ would exhibit stability, resulting in small values of $SD(n)$, $SD(T)$ and $SD(\hat{\alpha})$. Further, $\hat{\eta}$ should be near 1 and $\hat{\omega}$ near 0. The following tables contain the results of our simulations for $\alpha = 3$ (distribution skewed left), $\alpha = 1$ (symmetric) and $\alpha = 0.5$ (skewed right). Table I contains the results of the simulation for $\alpha = 3$. The first column contains the value of n^* , the second the value of the "delay" parameter γ and the remaining columns, the quantities defined above. Note that in the column labelled γ there are entries labelled "no delay" and that the larger γ the less the delay that is built into the procedure. The results in Table I indicate that for $\gamma = 1$ and 2 the procedure performs very well; $\bar{n}$ is close to n^* and the $n(i)$'s exhibit relatively little variation (albeit more than that exhibited by the $n(i)$'s in Mukhopadhyay, *et al.* (1983) wherein α was assumed to be known); the values of $\bar{T}$ are close to one, $SD(T)$ is small; $\hat{\eta}$ is near 1 and $\hat{\omega}$ is small. It is

clear that the performance procedure falls off as the delay parameter (γ) increases and that the results for $\gamma = 10$ are very similar to those wherein "no delay" was used. As $c \rightarrow 0$ and hence n^* increases, the performance of the procedure improves.

Tables II and III contain analogous information for the cases $\alpha = 1$ and 0,5, respectively. For choices of $\gamma = 0,5$ and 1, the results indicate that the procedure performs quite well. As in Mukhopadhyay, *et al.* (1983) it should be expected that we obtain better estimators for $\theta = 1$ for larger values of α . The results of the simulations reinforce this intuition. Overall, choices of γ near 1 seem to yield procedures which work well for all of the cases we considered.

As previously indicated, although the results of Theorem 1 have not been established for the "no delay" case, the results of the simulations clearly indicate the superior performance of the procedure for choices of γ of 0,5, 1 or 2.

4.2 Simulation results for the confidence interval procedure of Section 3

Again, our simulations were conducted for the power family of densities (see (1.1)) with $\theta = 1$ and $\alpha = 1/2, 1$ and 3. We focused on 95% confidence intervals ($\delta = 0,05$) of width $d = 0,10, 0,05, 0,01$ and 0,005 for values of the delay parameter $\gamma = 0,5, 1$ and 2. As indicated by (3.1), $n^{\#} = -(\alpha\gamma)^{-1} \ln \delta = 2,9957(\alpha\gamma)^{-1}$. In each instance we repeated the following procedure 1000 times. Sample from the indicated population until stopping rule (3.2) indicates that sampling be terminated. Record the i -th value of the sample size N , say $n(i)$ and determine whether $\theta (=1)$ is con-

tained in the interval $[T_{n(i)}, T_{n(i)} + d]$. Calculate $\bar{n} = k^{-1} \sum_{i=1}^k n(i)$.

$SD(n) = \{k^{-1} \sum_{i=1}^k (n(i) - \bar{n})^2\}^{1/2}$ for $k = 1000$ and the empirical coverage probability of the 1000 intervals generated. The results of our simulations are contained in the following tables. If the procedure performs well, we would expect $\bar{n}$ to be close to $n^{\#}$, $SD(n)$ to be relatively small and the empirical coverage probabilities to be near 0,95.

TABLE I: EMPIRICAL PERFORMANCE OF POINT ESTIMATION PROCEDURE OF SECTION 2; $\alpha = 3$

n^*	γ	$\bar{n}$	$SD(n)$	$\bar{T}$	$SD(T)$	$\bar{L}$	$\bar{\alpha}$	$SD(\hat{\alpha})$	$\hat{\eta}$	$\hat{\omega}$
5	NO DELAY	3,590	1,756	0,89856	0,10476	0,03396	32,48488	162,53557	1,27341	0,00729
	0,500	8,600	1,088	0,96127	0,03683	0,03336	4,20955	2,57616	1,25098	0,00669
	1,000	6,432	1,194	0,94792	0,05113	0,02812	5,42923	4,92077	1,05468	0,00146
	2,000	4,830	1,355	0,92753	0,07607	0,02814	9,14259	16,41350	1,05533	0,00148
	10,000	3,598	1,757	0,89856	0,10476	0,03399	32,46652	162,53844	1,27448	0,00732
25	NO DELAY	22,894	6,021	0,97970	0,03974	0,00258	26,03432	170,30243	2,41732	0,00151
	0,500	33,552	2,616	0,98976	0,01277	0,00116	3,26927	0,62446	1,08583	0,00009
	1,000	26,206	3,262	0,98706	0,01544	0,00109	3,40333	0,89110	1,01751	0,00002
	2,000	24,106	3,741	0,98553	0,01687	0,00111	3,47076	1,09051	1,04294	0,00005
	10,000	22,996	5,874	0,98027	0,03836	0,00245	25,43478	170,16171	2,29700	0,00138
100	NO DELAY	98,452	9,683	0,99611	0,01083	0,00011	38,15469	656,11035	1,66759	0,00004
	0,500	118,368	5,274	0,99710	0,00856	0,00007	3,07460	0,27121	1,01948	0,00000
	1,000	101,100	5,980	0,99653	0,00867	0,00007	3,08640	0,29314	1,01371	0,00000
	2,000	99,022	6,143	0,99649	0,00873	0,00007	3,08706	0,29837	1,00444	0,00000
	10,000	98,452	9,683	0,99611	0,01083	0,00011	38,15469	656,11035	1,66759	0,00004

TABLE II: EMPIRICAL PERFORMANCE OF POINT ESTIMATION PROCEDURE OF SECTION 2; $\alpha = 1$

n^*	γ	$\bar{n}$	$SD(n)$	$\bar{T}$	$SD(T)$	$\bar{L}$	$\bar{\alpha}$	$SD(\hat{\alpha})$	$\hat{\eta}$	$\hat{\omega}$
5	NO DELAY	3,044	1,465	0,70964	0,23808	0,23833	12,69769	43,54573	0,99305	-0,00167
	0,500	6,274	1,253	0,84826	0,13701	0,24250	1,94828	2,35267	1,01041	0,00250
	1,000	5,030	1,327	0,81270	0,16663	0,22374	3,61749	13,48805	0,93223	-0,01626
	2,000	3,830	1,478	0,75111	0,21722	0,23162	11,93500	43,67502	0,96510	-0,00838
	10,000	3,044	1,465	0,70964	0,23808	0,23833	12,69769	43,54573	0,99305	-0,00167
25	NO DELAY	21,132	7,719	0,90194	0,18279	0,04837	18,85530	184,79211	5,03871	0,03877
	0,500	28,904	3,046	0,96414	0,04375	0,01054	1,08950	0,23646	1,09741	0,00094
	1,000	24,846	3,488	0,95615	0,05155	0,01088	1,11863	0,27773	1,13288	0,00128
	2,000	23,556	4,028	0,95051	0,06451	0,01258	1,23811	1,98342	1,31011	0,00298
	10,000	21,234	7,641	0,90401	0,17972	0,04689	18,78062	184,79298	4,88389	0,03729
100	NO DELAY	97,194	15,119	0,97929	0,08652	0,00824	11,32162	101,03307	13,73170	0,00764
	0,500	108,770	6,193	0,99088	0,01230	0,00060	1,02119	0,10161	1,00589	0,00000
	1,000	100,160	6,560	0,99035	0,01262	0,00059	1,02180	0,10518	0,97815	-0,00001
	2,000	99,136	6,684	0,99028	0,01264	0,00058	1,02194	0,10617	0,97465	-0,00002
	10,000	97,410	14,517	0,98064	0,08140	0,00732	11,09149	100,92551	12,20813	0,00672

TABLE III: EMPIRICAL PERFORMANCE OF POINT ESTIMATION PROCEDURE OF SECTION 2; $\alpha = 0,5$.

n^*	γ	$\bar{n}$	SD(n)	$\bar{T}$	SD(T)	$\bar{L}$	$\bar{\alpha}$	SD($\hat{\alpha}$)	$\hat{\eta}$	$\hat{\omega}$
5	NO DELAY	2,756	1,243	0,53228	0,29687	0,65961	6,48623	22,04779	0,68710	-0,30039
	0,500	4,824	1,390	0,66338	0,25565	0,79603	1,98627	7,67703	0,82920	-0,16397
	1,000	3,818	1,373	0,59931	0,28831	0,73230	6,00196	22,10982	0,76281	-0,22770
	2,000	3,124	1,345	0,55600	0,30069	0,68736	6,43209	22,05930	0,71600	-0,27264
	10,000	2,758	1,242	0,53228	0,29687	0,65987	6,48705	22,04761	0,68736	-0,30013
25	NO DELAY	18,546	8,883	0,77618	0,28785	0,15188	5,46560	45,50270	3,95523	0,11348
	0,500	26,454	3,333	0,91968	0,08484	0,04067	0,55122	0,12865	1,05915	0,00227
	1,000	23,564	4,042	0,90414	0,10578	0,04444	0,57120	0,17852	1,15735	0,00604
	2,000	21,882	5,490	0,86895	0,17884	0,07150	0,68264	1,49732	1,86200	0,03310
	10,000	18,632	8,829	0,77827	0,28668	0,15037	5,42867	45,50092	3,91577	0,11197
100	NO DELAY	94,894	17,797	0,95452	0,14612	0,02488	3,50208	27,16728	10,36820	0,02248
	0,500	104,190	6,653	0,98041	0,02131	0,00245	0,51474	0,05486	1,01955	0,00005
	1,000	98,458	7,116	0,97920	0,02251	0,00246	0,51602	0,05751	1,02350	0,00006
	2,000	97,812	7,171	0,97903	0,02263	0,00246	0,51629	0,05769	1,02409	0,00006
	10,000	95,274	16,795	0,95768	0,13728	0,02211	3,37253	27,03418	9,21119	0,01971

Table IV contains the results of our simulations for $\gamma = 0.5$. The first column contains the value of α , the second the value of d , the third the empirical coverage probability (ECP) and the last three $n^\#$, $\bar{n}$ and $SD(n)$, respectively. For $\alpha = 0.5$, the procedure performs quite well; $\bar{n}$ is close to $n^\#$ and the empirical coverage probabilities are very near 0.95, when $\alpha = 1.0$, the procedure still performs quite well, tending to oversample modestly, resulting in coverage probabilities only slightly exceeding 0.95. For $\alpha = 3$, there is a tendency to "oversample" and the ECP's are high. For all choices of α , the performance of the procedure improves as $d \rightarrow 0$ and, hence, $n^\#$ increases.

Tables V and VI contain analogous results for $\gamma = 1$, and "no delay" respectively. An examination of Table V ($\gamma = 1$) shows that the procedure works very well with $\gamma = 1$: $\bar{n}$ is very close to $n^\#$ and the empirical coverage probabilities are uniformly close to 0.95 (except when $d = 0.1$). Again, the performance of the procedure improves as $d \rightarrow 0$. The same general comments pertain to the results we obtained when the procedure was simulated for $\gamma = 2$. Although we have omitted a table of the result to conserve space, the results indicate that except for the larger d values, the ECP's are very close to 0.95. However, the $\bar{n}$'s tend to be a little smaller than desired, indicating undersampling which may be compensated for by choosing a smaller delay parameter (i.e. further delaying the procedure).

Simultaneous examination of Tables IV, V and VI clearly demonstrates the superiority of the delayed procedures (Tables IV and V) over the procedure with no delay (Table VI). Overall, we would recommend use of the procedure with delay parameter, γ , near 1.

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TABLE IV: Empirical Performance of Sequential 95% Confidence Interval Procedure of Section 3; $\gamma = 0.5$.

α	d	ECP	$n^{\#}$	$\bar{n}$	SD(n)
0,500	0,100	0,949	59,915	60,331	7,985
0,500	0,050	0,954	119,829	121,831	10,910
0,500	0,010	0,935	599,146	608,625	23,059
0,500	0,005	0,938	1198,293	1212,020	33,171
1,000	0,100	0,952	29,957	33,277	4,894
1,000	0,050	0,965	59,915	65,413	7,158
1,000	0,010	0,973	299,573	314,908	16,272
1,000	0,005	0,958	599,146	621,225	22,819
3,000	0,100	0,993	9,986	16,846	1,984
3,000	0,050	0,989	19,972	30,285	3,153
3,000	0,010	0,973	99,858	125,602	8,309
3,000	0,005	0,972	199,716	237,664	12,278

TABLE V: Empirical Performance of Sequential 95% Confidence Interval Procedure of Section 3; $\gamma = 1.0$.

α	d	ECP	$n^{\#}$	$\bar{n}$	SD(n)
0,500	0,100	0,927	59,915	56,752	8,904
0,500	0,050	0,938	119,829	116,878	11,096
0,500	0,010	0,958	599,146	597,042	24,328
0,500	0,005	0,951	1198,293	1196,100	32,409
1,000	0,100	0,919	29,957	28,284	6,245
1,000	0,050	0,932	59,915	58,502	8,345
1,000	0,010	0,956	299,573	297,986	18,226
1,000	0,005	0,942	599,146	597,102	25,429
3,000	0,100	0,970	9,986	11,793	2,320
3,000	0,050	0,959	19,972	21,710	3,877
3,000	0,010	0,947	99,858	101,618	9,512
3,000	0,005	0,949	199,716	201,648	13,740

TABLE VI: Empirical Performance of Sequential 95% Confidence Interval Procedure of Section 3; NO DELAY.

α	d	ECP	$n^{\#}$	$\bar{n}$	SD(n)
0,500	0,100	0,738	59,915	44,734	23,567
0,500	0,050	0,825	119,829	103,076	38,299
0,500	0,010	0,915	599,146	578,204	104,077
0,500	0,005	0,934	1198,293	1176,603	153,160
1,000	0,100	0,689	29,957	20,501	12,486
1,000	0,050	0,798	59,915	49,001	20,972
1,000	0,010	0,930	299,573	289,042	50,115
1,000	0,005	0,934	599,146	590,858	61,186
3,000	0,100	0,693	9,986	5,840	4,081
3,000	0,050	0,754	19,972	14,264	8,278
3,000	0,010	0,915	99,858	94,546	21,604
3,000	0,005	0,922	199,716	193,774	32,970

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